Managing Users, Groups, and Sudo Privileges? SSH Security & Key-Based Authentication? Understanding SELinux & AppArmor

Q.1 Create a new user called streamer and a group called musicfans on your Linux system, then add streamer to the musicfans group. Verify the group membership using the id command.

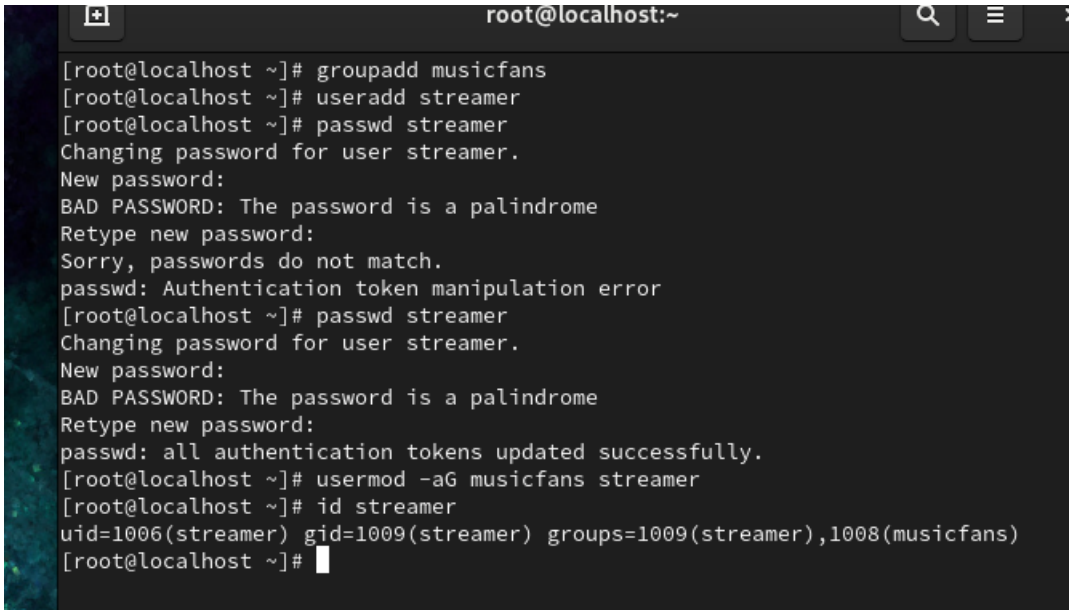
Ans- Step 1- groupadd musicfans

Step 2- useradd streamer

Step 3- passwd streamer

Step 4- usermod -aG musicfans streamer

Step 5- id streamer

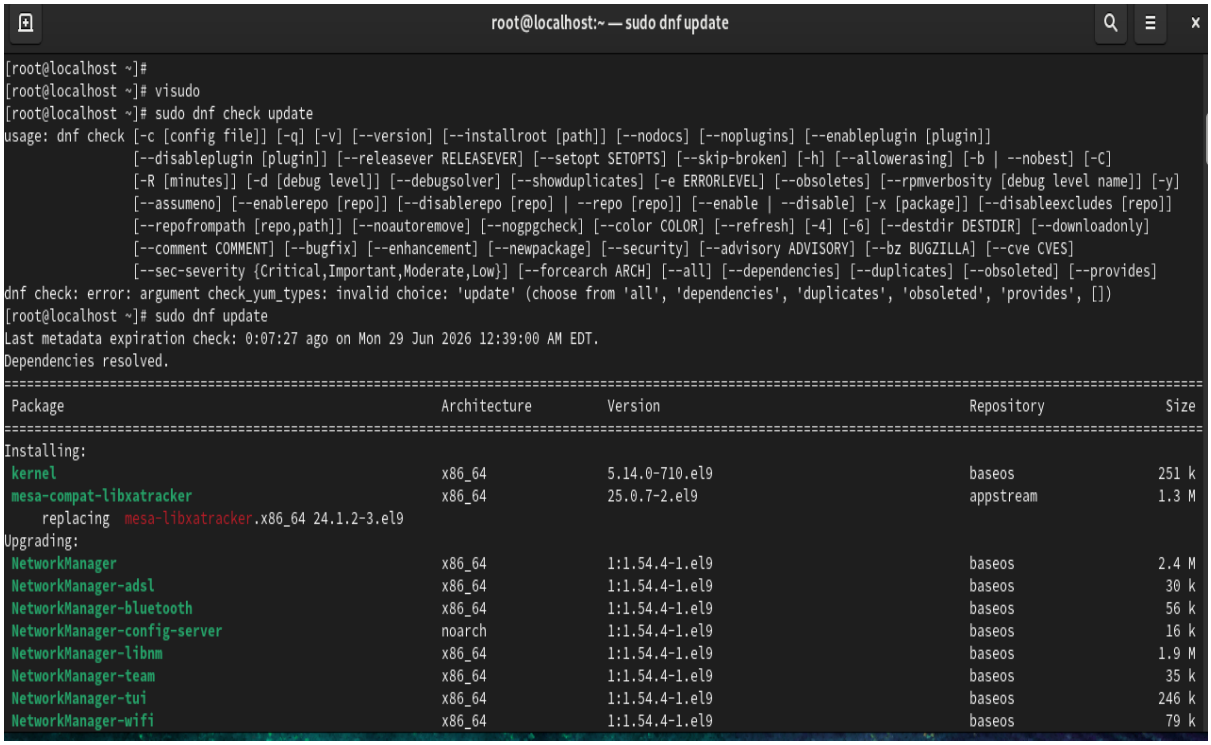


Q.2 Give your current user sudo privileges by editing the /etc/sudoers file using visudo. Test by running a command that requires sudo, such as updating the package list.

Ans- Step 1- visudo

Step 2- root ALL=(ALL)ALL

Step 3- Test: sudo dnf update



**Q.3 Set up SSH key-based authentication: generate an SSH key pair using ssh-keygen, then copy your public key to a remote Linux server (or a local VM) using ssh-copy-id. Disable password authentication for SSH in /etc/ssh/sshd_config and restart the SSH service.

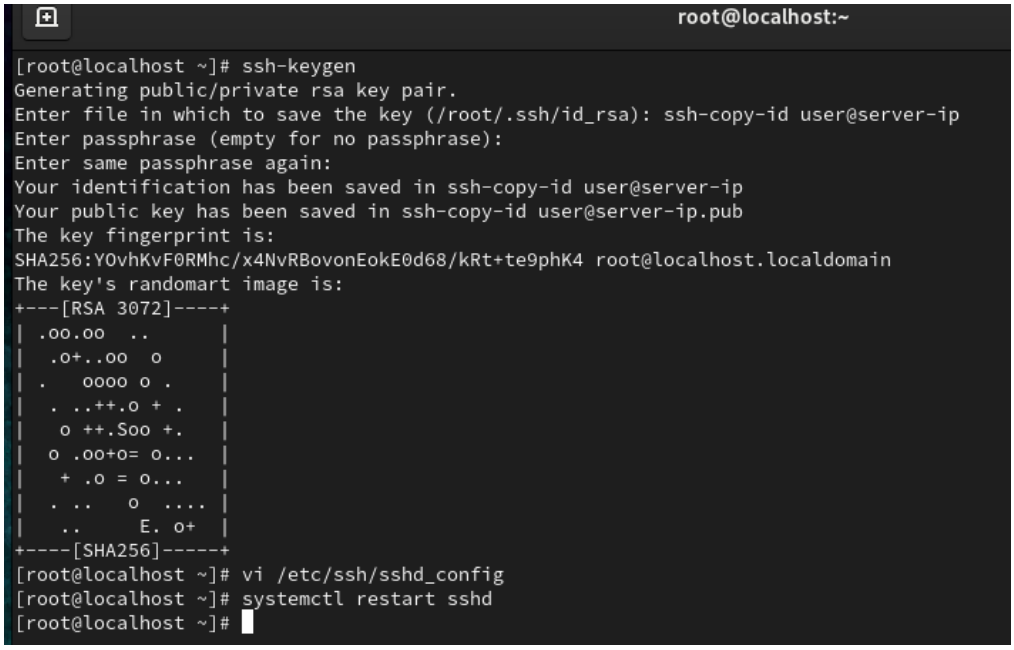
Hint: Look for the PasswordAuthentication setting in the SSH config file.**

Ans- Step 1-Generate SSH key: ssh-keygen

Step 2- Copy the public key: ssh-copy-id user@server-ip

Step 3- Edit SSH Configuration: vi/etc/ssh/shhd_config

Step 4-Restart SSH: systemctl restart sshd



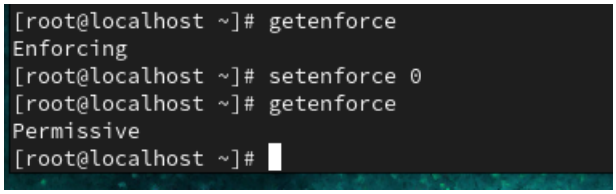
Q.4 Check if SELinux or AppArmor is enabled on your system. If SELinux is active, use the getenforce command to display its current mode, and switch it to permissive mode. If AppArmor is active, list all enforced profiles using sudo apparmor_status.

Ans- Step 1- Check status: getnforce

Step 2- Shows: Enforcing

Step 3- Change to permissive: setenforce 0

Step 4 - Verify: getenforce



Q.5 Use ChatGPT to explain the difference between SELinux and AppArmor in one paragraph, then write a summary in your own words and mention one real-world app (like Zomato or Paytm) that would benefit from these security modules.

Ans- SELinux and AppArmor are linux security modules that provide mandatory access control to protect the sytem. SELinux uses security labels and offers very detailed and strict security polocies,making it highly secure but more complex to configure. AppArmor uses file paths to define permissions, making it simpler to manage but lessgranular than SELinux. Both help prevent unauthorized access and limit the damage if an applicatiob is compromised.